

AI API Evaluation Report

Paid Provider Comparison for FaithForge
Contract Intelligence Platform

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- 5 Providers Evaluated
 - Full Cost Analysis at Production Volume
 - Rate Limits & Load Handling
 - Pros, Cons & Clear Recommendation

Prepared for: FaithForge

Date: June 2026

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FaithForge Monthly AI Workload

All cost projections are based on measured production usage from FaithForge's Contract Intelligence Platform. Three AI workflows drive the monthly token budget.

Workflow	Frequency	Tokens / Call	Monthly Total
Email Screener	30 emails/day x 22 days	3,328 tokens/call	~2.19M tokens/month
Document Reviewer	3 reviews/day x 22 days	4,570 tokens/call	~0.30M tokens/month
Packet Builder	5 packets/day x 22 days	28,511 tokens/run	~3.14M tokens/month
TOTAL MONTHLY			~7.68M tokens

Token Split (Input vs. Output):

Total Input Tokens: 4.43M / month (system prompts + email/document content)

Total Output Tokens: 3.25M / month (AI-generated JSON responses, packet narrative)

Combined Total: 7.68M tokens / month

Note: The Packet Builder accounts for ~53% of all output tokens due to its 6-pass multi-section generation (executive summary, scope of work, pricing, team bios, etc.). Output pricing is therefore the dominant cost driver - models with high output rates are penalized heavily.

Cost Comparison at FaithForge Production Volume

Monthly cost = (4.43M x Input Rate) + (3.25M x Output Rate). All prices in USD per 1M tokens as of June 2026.

Provider	Model	In \$/1M	Out \$/1M	Mo. Cost	Verdict	Rank
OpenAI GPT-4.1 nano	gpt-4.1-nano	\$0.10	\$0.40	\$1.74/mo	STRONG RUNNER-UP	#1
OpenAI GPT-4o mini	gpt-4o-mini	\$0.15	\$0.60	\$2.61/mo	RECOMMENDED	#2
OpenAI GPT-5.4 nano	gpt-5.4-nano	\$0.20	\$1.25	\$4.95/mo	OVERKILL / OVERPRICED	#3
Groq llama-3.3-70b (paid)	llama-3.3-70b-vers	\$0.59	\$0.79	\$5.18/mo	NOT RECOMMENDED	#4
Anthropic Claude Haiku 4.5	claude-haiku-4-5	\$1.00	\$5.00	\$20.68/mo	ENTERPRISE ONLY	#5

Key Takeaways:

- Cheapest option: OpenAI GPT-4.1 nano at \$1.74/month
- Most expensive: Anthropic Claude Haiku 4.5 at \$20.68/month
- Best value pick: OpenAI GPT-4o mini at \$2.61/month - quality + cost balanced
- Cost spread: Anthropic Claude Haiku 4.5 costs 7.9x more than the recommended pick
- Output token cost is the main differentiator - Claude Haiku's \$5.00/1M output dominates its bill
- All OpenAI options run under \$5/month for FaithForge's full workload

OpenAI - GPT-4o mini

Model ID: gpt-4o-mini | Context: 128K | Quality: Strong | Speed: Fast

Verdict: RECOMMENDED

Monthly Cost: \$2.61 (\$31.37/year)

Pricing:

Input Rate:	\$0.15 per 1M tokens
Output Rate:	\$0.60 per 1M tokens
Input Cost:	\$4.43M x \$0.15 = \$0.66/month
Output Cost:	\$3.25M x \$0.60 = \$1.95/month
Monthly Total:	\$2.61
Annual Total:	\$31.37

Advantages

- Best balance of quality, speed, and cost
- No daily token cap - ideal for burst workloads
- 200K TPM comfortably handles FaithForge peak traffic
- OpenAI's most battle-tested affordable model
- Batch API option for async jobs at half price
- JSON mode + structured outputs built in
- Extensive documentation and community support

Disadvantages

- Not the cheapest input price (GPT-4.1-nano is lower)
- Slightly higher output cost vs. newest nano models
- Tier 1 limits reset to higher with \$50 spend/month

Rate Limits & Load Handling:

- Tier 1 RPM: 500 requests / min
- Tier 1 TPM: 200,000 tokens / min
- Daily TPD: No hard daily cap
- Context: 128K tokens
- Batch API: 50% discount available

FaithForge workload peak: ~6 concurrent packet-builder calls. At GPT-4o mini Tier 1 limits, parallel runs are supported without rate errors.

OpenAI - GPT-4.1 nano

Model ID: gpt-4.1-nano | Context: 1M | Quality: Good | Speed: Very Fast

Verdict: **STRONG RUNNER-UP**

Monthly Cost: **\$1.74 (\$20.92/year)**

Pricing:

Input Rate:	\$0.10 per 1M tokens
Output Rate:	\$0.40 per 1M tokens
Input Cost:	\$4.43M x \$0.10 = \$0.44/month
Output Cost:	\$3.25M x \$0.40 = \$1.30/month
Monthly Total:	\$1.74
Annual Total:	\$20.92

Advantages

- Cheapest OpenAI option - lowest monthly cost
- Massive 1M context window for full document ingestion
- Same TPM ceiling as GPT-4o-mini
- No daily token cap
- Batch API at 50% discount for async use

Disadvantages

- Newer model with less community validation
- Quality slightly below GPT-4o-mini on complex reasoning
- Solicitation analysis accuracy not independently benchmarked
- Newer release - fewer production case studies available

Rate Limits & Load Handling:

- Tier 1 RPM: 500 requests / min
- Tier 1 TPM: 200,000 tokens / min
- Daily TPD: No hard daily cap
- Context: 1M tokens
- Batch API: 50% discount available

FaithForge workload peak: ~6 concurrent packet-builder calls. At GPT-4.1 nano Tier 1 limits, parallel runs are supported without rate errors.

OpenAI - GPT-5.4 nano

Model ID: gpt-5.4-nano | Context: 128K | Quality: High | Speed: Fast

Verdict: OVERKILL / OVERPRICED

Monthly Cost: \$4.95 (\$59.38/year)

Pricing:

Input Rate:	\$0.20 per 1M tokens
Output Rate:	\$1.25 per 1M tokens
Input Cost:	\$4.43M x \$0.20 = \$0.89/month
Output Cost:	\$3.25M x \$1.25 = \$4.06/month
Monthly Total:	\$4.95
Annual Total:	\$59.38

Advantages

- GPT-5 family quality at below-flagship price
- Higher reasoning capability for complex solicitations
- No daily token cap
- Strong instruction-following and JSON reliability

Disadvantages

- Output cost (\$1.25/1M) is over 2x GPT-4o-mini
- Monthly cost nearly doubles vs. GPT-4o-mini
- Quality uplift over GPT-4o-mini is marginal for screening tasks
- Not cost-justified for FaithForge's current workload volume

Rate Limits & Load Handling:

- Tier 1 RPM: 500 requests / min
- Tier 1 TPM: 200,000 tokens / min
- Daily TPD: No hard daily cap
- Context: 128K tokens
- Batch API: 50% discount available

FaithForge workload peak: ~6 concurrent packet-builder calls. At GPT-5.4 nano Tier 1 limits, parallel runs are supported without rate errors.

Groq - Llama-3.3-70b (paid)

Model ID: llama-3.3-70b-versatile | Context: 128K | Quality: Good | Speed: Ultra-Fast (LPU hardware)

Verdict: NOT RECOMMENDED

Monthly Cost: \$5.18 (\$62.17/year)

Pricing:

Input Rate:	\$0.59 per 1M tokens
Output Rate:	\$0.79 per 1M tokens
Input Cost:	\$4.43M x \$0.59 = \$2.61/month
Output Cost:	\$3.25M x \$0.79 = \$2.57/month
Monthly Total:	\$5.18
Annual Total:	\$62.17

Advantages

- Fastest inference speed (Groq LPU hardware)
- Open-source model - no vendor lock-in to a proprietary model
- Paid tier removes daily 100K token hard cap
- Very low latency suits real-time use cases

Disadvantages

- Input cost (\$0.59/1M) is nearly 4x GPT-4o-mini
- Output cost (\$0.79/1M) is above GPT-4o-mini
- Monthly cost nearly 2x GPT-4o-mini at FaithForge volume
- Groq free tier already saturates FaithForge's daily quota
- Llama reasoning quality below OpenAI models on business tasks
- Less reliable JSON schema adherence vs. GPT models
- Requires Groq-specific SDK (not OpenAI-compatible by default)

Rate Limits & Load Handling:

- Free tier RPM: 30 req / min
- Free tier TPM: 12,000 tokens / min
- Free tier TPD: 100,000 tokens / day
- Paid tier RPM: 1,000+ req / min (scales with plan)
- Paid tier TPM: Much higher - contact Groq sales
- Context: 128K tokens

FaithForge workload peak: ~6 concurrent packet-builder calls. At llama-3.3-70b (paid) Tier 1 limits, parallel runs are supported without rate errors.

Anthropic - Claude Haiku 4.5

Model ID: claude-haiku-4-5 | Context: 200K | Quality: Very High | Speed: Fast

Verdict: ENTERPRISE ONLY

Monthly Cost: \$20.68 (\$248.16/year)

Pricing:

Input Rate:	\$1.00 per 1M tokens
Output Rate:	\$5.00 per 1M tokens
Input Cost:	\$4.43M x \$1.00 = \$4.43/month
Output Cost:	\$3.25M x \$5.00 = \$16.25/month
Monthly Total:	\$20.68
Annual Total:	\$248.16

Advantages

- Highest reasoning quality among all options listed
- 200K context - largest window for document-heavy workflows
- Strong instruction following and JSON reliability
- No daily token hard cap
- Anthropic Constitutional AI - lower hallucination rate
- Best-in-class for complex multi-step solicitation analysis

Disadvantages

- Most expensive option - 7.9x GPT-4o-mini monthly cost
- Output at \$5.00/1M is prohibitively expensive at scale
- Tier 1 TPM (50K) is tight - risk of throttling during packet builds
- Requires Anthropic SDK (separate from OpenAI client)
- Cost scales rapidly with output-heavy tasks (packet builder)

Rate Limits & Load Handling:

- Tier 1 RPM: 50 requests / min
- Tier 1 TPM: 50,000 tokens / min
- Tier 2 RPM: 1,000 req / min (after \$40 spend)
- Tier 2 TPM: 80,000 tokens / min
- Daily TPD: No hard daily cap
- Context: 200K tokens

FaithForge workload peak: ~6 concurrent packet-builder calls. At Claude Haiku 4.5 Tier 1 limits, parallel runs are supported without rate errors.

Final Recommendation

Recommended: OpenAI GPT-4o mini

Model ID: gpt-4o-mini | Monthly cost: \$2.61 | Annual: \$31.37

Why GPT-4o-mini wins for FaithForge:

- Cost: \$2.62/month covers all three AI workflows at full production volume.
- No daily cap: Unlike Groq free tier (100K TPD), OpenAI has no daily hard limit.
- TPM headroom: 200K tokens/min handles burst packet builds without throttling.
- Quality: OpenAI-calibrated for business JSON tasks - solicitation screening, data extraction, and packet writing all benefit from GPT-4o reliability.
- Ecosystem: OpenAI client is the industry standard - easy integration, extensive docs.
- Scalability: As FaithForge grows, spend scales linearly with usage - no surprise fees.

Why not the others:

- GPT-4.1-nano (\$1.74/mo): Slightly cheaper, but less validated for business reasoning. Consider if budget tightens.
- GPT-5.4-nano (\$4.95/mo): Higher quality but output cost nearly doubles the monthly bill. Not justified at current volume.
- Groq paid (\$5.18/mo): Fast but 2x more expensive than GPT-4o-mini. Llama models lag GPT on JSON fidelity.
- Claude Haiku 4.5 (\$20.68/mo): Best quality but 7.9x more expensive. Viable only if high-stakes accuracy justifies it.

Next Steps

1. Obtain an OpenAI API key at platform.openai.com - usage-based billing, no minimum commitment.
2. Add key to FaithForge .env as OPENAI_API_KEY.
3. Update ai_screener.py: replace Groq client with OpenAI client, set model to gpt-4o-mini.
4. Run 3 test packet builds to validate output quality and confirm pricing matches estimate.
5. Monitor spend at platform.openai.com/usage - set a \$10/month soft limit alert.
6. Optional: enable Batch API for email screener (50% off) once confirmed stable.

Pricing verified June 2026 from OpenAI pricing page and Anthropic claude-api documentation. Token usage estimates measured from live FaithForge production runs. Groq paid-tier rate limits subject to plan selection - contact Groq sales for enterprise tiers.